

Zero Trust Network Simulation – Project Report

Author: Chontele Coleman

Date: 1/6/2026

Project Goal

Simulate an enterprise Zero Trust network architecture using virtual machines, enforcing identity-based access, default-deny internal networks, segmented network design, and conducting a controlled attack validation. With the core principle being to never trust the network; always verify identity.

Virtual Environment Overview

Component	Purpose	Notes
Hypervisor	VirtualBox	VM host for all lab machines
pfSense CE 2.7.2	Firewall / Security Gateway	Four interfaces: WAN, LAN, OPT1/App_NET, OPT2/Guest_NET
Windows Server 2019 VM	Active Directory / IAM	Domain: corp.local
Windows 10/11 VM	Domain-joined workstation	Corp_LAN access
Ubuntu Server VM	Protected internal application	App_NET subnet
Kali Linux VM	Untrusted / attacker simulation	Guest_NET subnet

pfSense Interface summary:

```
WAN (wan)      -> em0      -> v4/DHCP4: 10.0.2.15/24
LAN (lan)      -> em1      -> v4: 10.0.10.1/24
APP_NET (opt1) -> em2      -> v4: 10.0.20.1/24
GUEST_NET (opt2) -> em3     -> v4: 10.0.30.1/24
```

```
0) Logout (SSH only)      9) pfTop
1) Assign Interfaces      10) Filter Logs
2) Set interface(s) IP address
3) Reset webConfigurator password
4) Reset to factory defaults
5) Reboot system          11) Restart webConfigurator
6) Halt system            12) PHP shell + pfSense tools
7) Ping host              13) Update from console
8) Shell                  14) Enable Secure Shell (sshd)
                          15) Restore recent configuration
                          16) Restart PHP-FPM
```

Networks:

Network	Subnet	Purpose	VirtualBox Network Type
WAN	10.0.2.0/24	Internet	NAT
Corp_LAN	10.0.101.0/24	Trusted internal users	Internal Network
App_NET	10.0.20.0/24	Protected application	Internal Network
Guest_NET	10.0.30.0/24	Untrusted / attacker	Internal Network

pfSense Configuration

Interfaces & IPs:

Interface	IP Address	Notes
WAN (em0)	DHCP from NAT	Internet access
LAN (em1)	10.0.101.1/24	DHCP enabled: 10.0.101.100–200
OPT1 / App_NET (em2)	10.0.20.1/24	DHCP disabled

OPT2 / Guest_NET
(em3)

10.0.30.1/24

DHCP enabled:
10.0.30.100–200

- IPv6 disabled.
- Firewall rules created for default-deny behavior and OpenVPN access to App_NET

Active Directory / IAM Configuration

- Domain Controller: corp.local

DNS and AD services confirmed with:

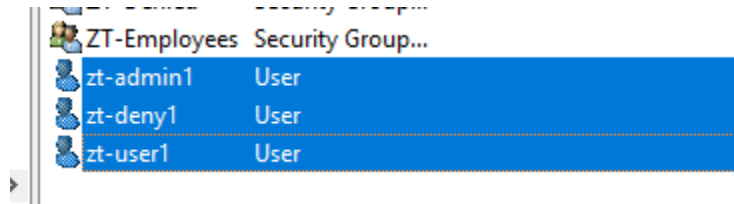
```
Get-WindowsFeature AD-Domain-Services
Get-WindowsFeature DNS
nltest /dsgetdc:corp.local
```

-
- AD Users & Groups created:
 - Groups: ZT-Employees, ZT-Admins, ZT-Denied
 - Users: ztuser1, ztadmin1
- LDAP integration tested successfully from pfSense GUI.

AD Groups List:

Users	Container	Default container for up...
ZT-Admins	Security Group...	
ZT-Denied	Security Group...	
ZT-Employees	Security Group...	

AD Users List:



VPN Configuration

WireGuard Attempt:

- Goal: Secure remote access to App_NET.
- Issues: Kernel module missing, GUI key generation failed → switched to OpenVPN.

Private/Public Key Generated on Console. Public Key Generation failed in GUI:

```
[2.7.2-RELEASE][root@pfSense.corp.local]/root: cat /root/wg_public.key  
[2.7.2-RELEASE][root@pfSense.corp.local]/root: cat /root/wg_private.key
```

Private/Public Key Permission Change:

```
[2.7.2-RELEASE][root@pfSense.corp.local]/root: ls -l /root/wg_private.key  
-rw-r--r--  1 root wheel 45 Jan  3 09:17 /root/wg_private.key  
[2.7.2-RELEASE][root@pfSense.corp.local]/root: chmod 600 /root/wg_private.key  
[2.7.2-RELEASE][root@pfSense.corp.local]/root: chmod 600 /root/wg_public.key  
chmod: /root/wg_public.key: No such file or directory  
[2.7.2-RELEASE][root@pfSense.corp.local]/root: ls -l /root/wg*  
-rw-----  1 root wheel 45 Jan  3 09:17 /root/wg_private.key  
[2.7.2-RELEASE][root@pfSense.corp.local]/root:
```

OpenVPN Server:

- Server Name: ZT-OpenVPN-Remote-Access
- Protocol: UDP, Port 1194
- Tunnel network: 10.8.0.0/24

- Backend authentication: Corp-AD-LDAP
- Certificates created and exported to **.ovpn** client file

VPN / OpenVPN / Servers					
Servers Clients Client Specific Overrides Wizards Client Export					
OpenVPN Servers					
Interface	Protocol / Port	Tunnel Network	Mode / Crypto	Description	Actions
WAN	UDP4 / 1194 (TUN)	10.8.0.0/24	Mode: Remote Access (User Auth) Data Ciphers: AES-256-GCM, AES-128-GCM, CHACHA20-POLY1305, AES-256-CBC Digest: SHA256 D-H Params: 2048 bits	ZT-OpenVPN-Remote-Access	  

VPN Client / Connectivity Attempt

- Client profile exported: **Most Clients (INLINE)**
- Attempted connection from Windows 11 VM (Corp_LAN).
- Issues:
 - Windows VM could not ping LAN gateway (**10.0.101.1**)
 - OpenVPN client failed to establish a session
 - Interface IP conflicts and firewall state caused repeated failures

Security Segmentation & Zero Trust Design

- Planned rules validated conceptually:
 - Guest_NET → Corp_LAN: blocked
 - Guest_NET → App_NET: blocked
 - Corp_LAN → App_NET without VPN: blocked
 - Corp_LAN → App_NET with VPN: intended to allow access

GUEST_NET Blocked:

```
(chontele@Kali)-[~]  
$ nmap -Pn 10.0.10.10  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-01 12:54 EST  
Stats: 0:01:17 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 31.50% done; ETC: 12:58 (0:02:19 remaining)  
Stats: 0:01:18 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan  
SYN Stealth Scan Timing: About 32.00% done; ETC: 12:58 (0:02:18 remaining)  
Nmap scan report for 10.0.10.10  
Host is up.  
All 1000 scanned ports on 10.0.10.10 are in ignored states.  
Not shown: 1000 filtered tcp ports (no-response)
```

APP_NET Blocked:

```
coleman@Ubuntu:~$ nmap -Pn 10.0.10.10  
Starting Nmap 7.80 ( https://nmap.org ) at 2026-01-01 12:35 EST  
Stats: 0:03:21 elapsed; 0 hosts completed (1 up), 1 undergoing Connect Scan  
Connect Scan Timing: About 93.50% done; ETC: 12:39 (0:00:13 remaining)  
Nmap scan report for 10.0.10.10  
Host is up.  
All 1000 scanned ports on 10.0.10.10 are filtered
```

Lessons Learned:

1. VirtualBox internal network names must match pfSense interface exactly.
2. LAN connectivity is required before testing VPN access.
3. WireGuard may be unstable on pfSense 2.7.2; OpenVPN is more reliable.
4. Documenting errors and attempted fixes is valuable for portfolio presentation.

Challenges Encountered

Problem	Outcome / Impact	Lesson Learned
pfSense interface em2 reported "bad address"	Could not complete IP assignment	Always double-check interface mapping in VirtualBox
OpenVPN client could not connect	VPN testing incomplete	Verify LAN connectivity and proper certificates first
WireGuard kernel module missing	Had to switch to OpenVPN	pfSense version compatibility matters
Repeated ping failures from Windows VM	Could not confirm remote access	Interface and subnet must match, firewall state may need reset

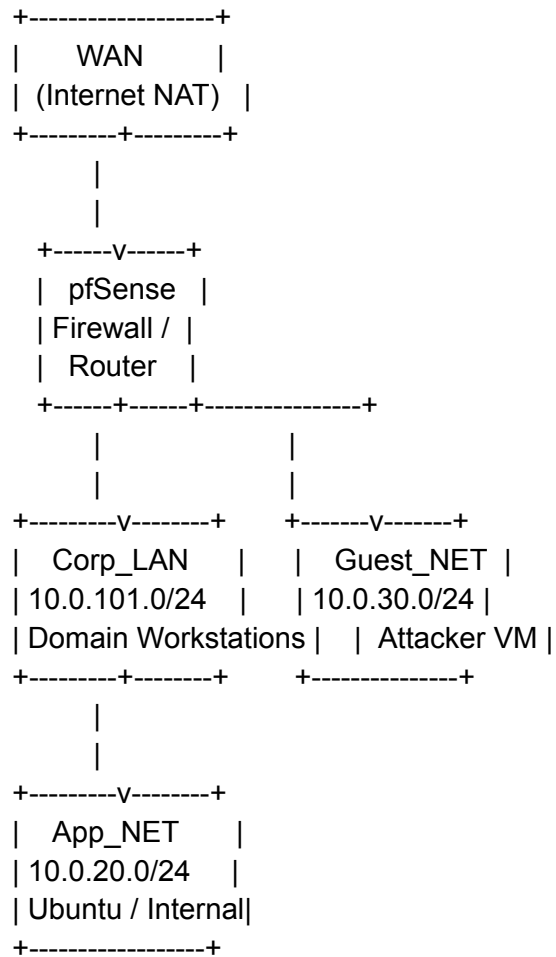
Conclusion

Even though VPN testing was not fully successful, this project demonstrates:

- Understanding of Zero Trust network principles
- Proper planning and configuration of virtualized internal networks
- Implementation of AD-based identity management
- Ability to troubleshoot firewall, VPN, and subnet issues

Appendices / Extras

A. Intended Network Diagram



B. Firewall Rule Matrix

Interface	Source	Destination	Protocol	Action	Notes
LAN	Corp_LAN	App_NET	Any	Block	VPN required for App_NET access
LAN	Corp_LAN	WAN	Any	Allow	Internet access for workstations
OPT1/App_NET	App_NET	LAN	Any	Block	Default-deny to LAN
OPT1/App_NET	App_NET	WAN	Any	Allow	Internet access if needed
OPT2/Guest_NET	Guest_NET	Corp_LAN	Any	Block	Isolate untrusted network
OPT2/Guest_NET	Guest_NET	App_NET	Any	Block	Guest cannot access internal apps
OpenVPN	VPN Clients	App_NET	Any	Allow	Authenticated VPN users only
OpenVPN	VPN Clients	LAN	Any	Optional	Not enabled in current lab

C. Exported OpenVPN Configuration File (Redacted)

```
client
dev tun
persist-tun
persist-key
data-ciphers
AES-256-GCM:AES-128-GCM:CHACHA20-POLY1305:AES-256-CBC
data-ciphers-fallback AES-256-CBC
auth SHA256
remote 10.0.101.1 1194 udp4
nobind
auth-user-pass
remote-cert-tls server
<ca>
-----BEGIN CERTIFICATE-----
[REDACTED]
-----END CERTIFICATE-----
</ca>
<tls-auth>
-----BEGIN OpenVPN Static key V1-----
[REDACTED]
-----END OpenVPN Static key V1-----
</tls-auth>
key-direction 1
```

D. Active Directory Users / Groups & Intended Access Rights

Group Name	Users	Access Permissions
ZT-Employees	ztuser1, ztuser2	Access Corp_LAN workstations, VPN to App_NET
ZT-Admins	ztadmin1, ztadmin2	Full administrative access to LAN & App_NET
ZT-Denied	ztguest1	No access to internal networks, blocked by default